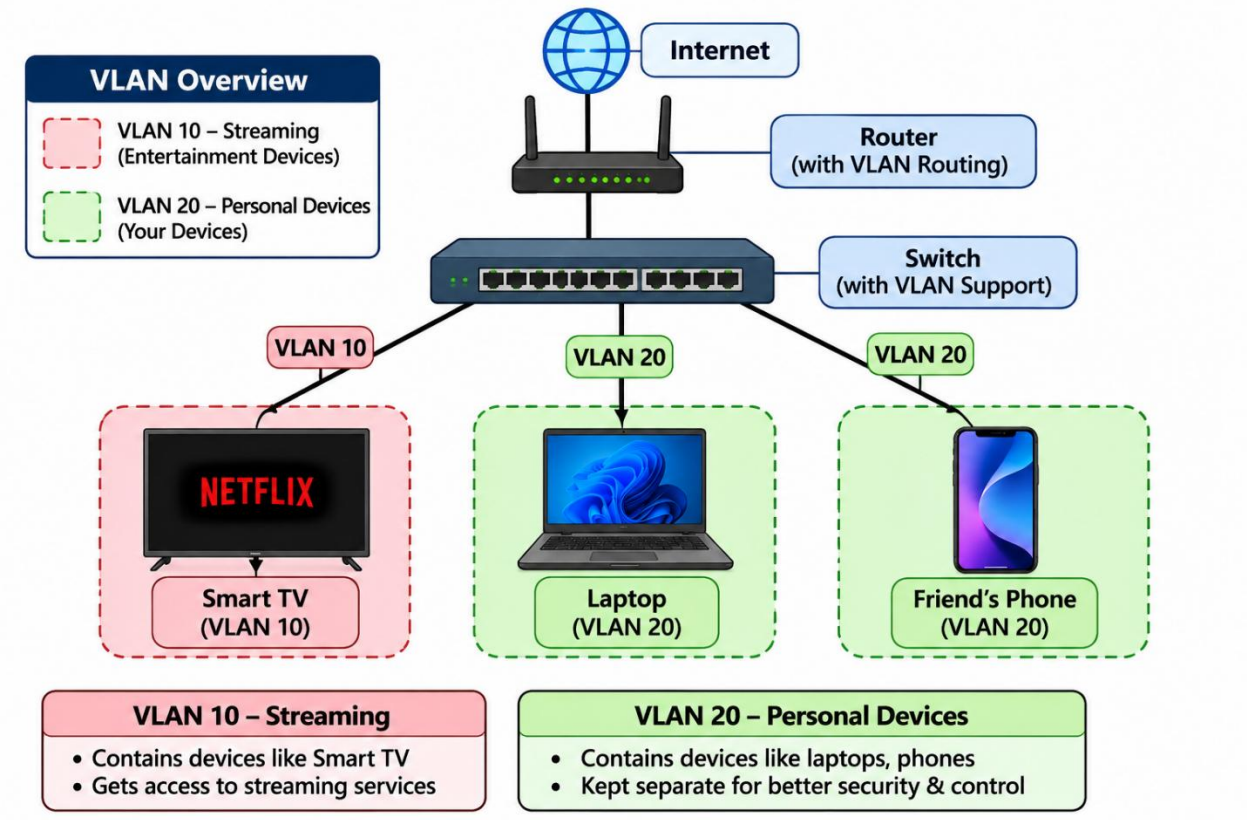
Switching and VLANS

Tasks 1 : Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



Explanation

A managed switch connects all three devices. VLAN 10 is assigned to streaming devices, while VLAN 20 is assigned to personal devices. VLANs create separate logical networks on the same physical switch and help improve network organization and isolation.

Task 2 : Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

Task 2: Configure and Verify VLANs in Cisco Packet Tracer

Objective

To create VLAN 10 and VLAN 20, assign five PCs to the VLANs, and verify communication between devices.

Configuration

- Created VLAN 10 named STREAMING.
- Created VLAN 20 named PERSONAL.
- Assigned PC0, PC1, and PC2 to VLAN 10.
- Assigned PC3 and PC4 to VLAN 20.
- Configured IP addresses in separate subnets.

Verification

The show vlan brief command was used to verify VLAN membership.

- PCs in the same VLAN were expected to communicate successfully.
- PCs in different VLANs were expected to fail to communicate because inter-VLAN routing was not configured.

Result

The switch was configured with two VLANs, and five PCs were assigned to the appropriate VLANs.

Command :

enable

configure terminal

vlan 10

name STREAMING

exit

```
vlan 20
name PERSONAL
exit
interface range fastEthernet 0/1-3
switchport mode access
switchport access vlan 10
exit
interface range fastEthernet 0/4-5
switchport mode access
switchport access vlan 20
exit
end
show vlan brief
```

Task 3 : In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

To move a PC from VLAN 10 to VLAN 20 and verify the connectivity change.

Before Configuration

PC0 was connected to switch port Fa0/1, which was assigned to VLAN 10.

interface fastEthernet 0/1

switchport mode access

switchport access vlan 10

After Configuration

The switch port was reassigned to VLAN 20.

configure terminal

interface fastEthernet 0/1

switchport mode access

switchport access vlan 20

end

show vlan brief

PC0's IP address was changed to 192.168.20.3 with subnet mask 255.255.255.0.

Connectivity Testing

- Ping to a PC in VLAN 20 was expected to succeed.
- Ping to a PC in VLAN 10 was expected to fail because inter-VLAN routing was not configured.

Result

PC0 was moved from VLAN 10 to VLAN 20 by changing its switch-port assignment and IP address.

Task 4 : Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3.

Task 4: Automating VLAN Assignment for 10 Switch Ports

Objective

To assign ten switch ports to VLAN 30 using a range command in Cisco Packet Tracer.

Commands Used

enable

configure terminal

vlan 30

name VLAN30_USERS

exit

interface range fastEthernet 0/1-10

switchport mode access

switchport access vlan 30

exit

end

show vlan brief

Explanation

The vlan 30 command creates VLAN 30. The interface range fastEthernet 0/1-10 command selects ten FastEthernet ports simultaneously. The switchport access vlan 30 command assigns all selected ports to VLAN 30.

Result

Ten switch ports were assigned to VLAN 30 using a single interface range command.